

Asymptotic Calibration of the BFS Window and Effective Dimension of the Admissible Trajectory

O28 — Spectral Admissibility Sub-programme

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Abstract

The O25 programme [8] identified the ratio $n_1(q)/q$ as the central open asymptotic variable controlling the finite-size drift of $\bar{\delta}_{\text{pair}}(q)$, and O26–O27 [6, 7] identified the effective dimension $r_{\text{eff}} = d_\rho^2$ of the per-pair covariance operator in $\text{End}(V_\rho)$ as the key falsifiability criterion for the representation-theoretic identification. The present paper reports two measurements from the Q5a-O5 checkpoints for $q \in \{29, 61, 101, 151\}$. First, we extract $n_1(q)$ from the auto-calibrated fitting windows and fit $n_1(q) = \hat{\alpha}q + \hat{\beta}$ by ordinary least squares, obtaining $\hat{\alpha} \approx 0.053$ with $R^2 \approx 0.879$. The O14 correction [4] yields $\delta_{\text{corr}}(q) \in [7.4, 10.6]$ for all $q \in \{29, 61, 101, 151, 211\}$, confirming the structural conclusion of O25. Second, using the per-block Weil vector projections $\pi_c(v)$ stored in the Q5a-O5 checkpoints, we perform the formal effective-dimension computation of O26 Criterion 5.4 in $\text{End}(H_{\text{eff}})$, where $H_{\text{eff}} = \mathbb{C}^3$ is the admissible projection space ($\text{HEFF.DIM} = 3$, consistent with $\Sigma_c(n_3) = 3$ from O23 [9]). The covariance $\mathcal{C}_c$ has rank exactly $r_{\text{eff}} = 3$ for every conjugate pair and every prime $q \in \{61, 101, 151\}$ (100% of pairs), with normalised eigenvalue structure $[\lambda_1 : \lambda_2 : \lambda_3] = [1 : \frac{1}{2} : \frac{1}{2}]$ invariant across all pairs and primes. This rank-3 result fills H_{eff} completely. The gap from the spin- $\frac{1}{2}$ prediction $d_\rho^2 = 4$ reflects that $H_{\text{eff}} \neq V_\rho$: the formal test within $\text{End}(V_\rho)$ requires first identifying $V_\rho \subset H_{\text{eff}}$, which is the object of O29.

Keywords. spectral admissibility; BFS window depth; pair observable; Weil representation; Heisenberg graphs; capacity exponent; normalisation correction; covariance operator; effective dimension; admissible trajectory; Born–Infeld saturation

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1 Position of the Problem

1.1 What O25–O27 established

Paper O25 [8] established three results for the pair observable $\sigma_{\text{pair}}(n) = \sigma_c(n) \sigma_{q-c}(n)$ on $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$: (i) the inter-pair standard deviation of δ_{pair} decreases monotonically from 0.54 at $q = 29$ to 0.11 at $q = 211$, confirming that δ_{pair} is a structural invariant of the Weil representation; (ii) empirical fits carry no asymptotic meaning because the leading finite-size correction $\eta \log q / \log n_1(q) \approx 1$ is itself q -dependent; (iii) when the O14 correction is applied, the corrected values fall within the admissible window [7.4, 10.6]. O25 identified the ratio $n_1(q)/q$ as the central open direction.

Paper O26 [6] established a dictionary between conjugate Weil blocks and rank-one matrix coefficients in a representation space, and formulated a hierarchy of identifications (Levels I–III). Test 4 of O26 (Criterion 5.4) provides the key falsifiability criterion: the effective dimension r_{eff} of the per-pair covariance operator

$$\mathcal{C}_c = \frac{1}{N} \sum_{n=n_0}^{n_1} \text{vec}(\widetilde{M}_n^f) \text{vec}(\widetilde{M}_n^f)^\dagger \quad (1)$$

in $\text{End}(V_\rho)$ should equal $d_\rho^2 = 4$ for the spin- $\frac{1}{2}$ sector candidate.

Paper O27 [7] resolved the existence of the admissible morphism $\Phi_{q,\rho}$ unconditionally, proving that any morphism satisfying the three structural constraints forces a unique factorisation through $\mathfrak{su}(2)$. The full implication chain

$$\text{emergence} \Rightarrow \text{non-injectivity [3]} \Rightarrow \text{pair structure} \Rightarrow \text{quadratic form} \Rightarrow \mathfrak{su}(2)$$

is thereby closed at the structural level. The remaining open question after O27 is the numerical identification of the correct irreducible sector from the covariance structure of the admissible trajectory.

1.2 Scope of O28

O28 addresses the two numerical questions left open after O25–O27:

1. **Part A:** calibrate the asymptotic constant $\hat{\alpha}$ in $n_1(q) \sim \hat{\alpha} q$ and reassess $\delta_{\text{corr}}(q)$ using the measured n_1 values.
2. **Part B:** perform the formal effective-dimension computation of O26 Criterion 5.4 using the per-block Weil vector projections $\pi_c(v) \in H_{\text{eff}}$ stored in the Q5a-O5 checkpoints, and determine what the covariance structure of $\mathcal{C}_c$ in $\text{End}(H_{\text{eff}})$ reveals about the irreducible sector.

2 Setup and Notation

We follow the notation of O25 [8]. Let q be an odd prime, $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, and $\text{Cay}(G_q, S_q)$ its Cayley graph with generating set $S_q = \{X^{\pm 1}, Y^{\pm 1}\}$. The pair observable is

$$\sigma_c(n) = \frac{\delta r_n}{|S_n|}, \quad \sigma_{\text{pair}}(n) = \sigma_c(n) \sigma_{q-c}(n).$$

The fitting window $[n_0, n_1]$ is auto-calibrated from the BFS data via the `find_fitting_window` procedure of O12 [1], with `bfs_frac` = 0.99. The O14 normalisation correction [4] gives

$$\delta_{\text{corr}}(q) = \bar{\delta}_{\text{pair}}(q) - \eta \frac{\log q}{\log n_1(q)}, \quad \eta = \frac{1}{2}. \quad (2)$$

The Q5a-O5 pipeline additionally stores, for each conjugate pair $(c, q - c)$ and each window shell s , the per-vector H_{eff} -projections

$$\pi_c(v_j^{(s)}) = B_{\text{eff}} v_j^{(s)} \in H_{\text{eff}} = \mathbb{C}^3, \quad (3)$$

where $B_{\text{eff}} \in \mathbb{C}^{3 \times q}$ is the first $\text{HEFF_DIM} = 3$ rows of the Gram–Schmidt basis matrix accumulated up to shell n_1 , and $v_j^{(s)} \in \mathbb{C}^q$ is the j -th Weil fingerprint vector at shell s .

3 Part A: Asymptotic Calibration of $n_1(q)/q$

3.1 Extraction of $n_1(q)$ from checkpoints

For each prime $q \in \{29, 61, 101, 151\}$, the value $n_1(q)$ is extracted from the `n1` key of the Q5a-O5 checkpoint. The value for $q = 211$ is taken from the O25 checkpoint [8]. Table 1 collects the complete dataset.

Table 1: BFS window depth and asymptotic ratio. $\bar{\delta}_{\text{pair}}(q)$: mean of δ_{pair} over all conjugate pairs. $\eta \log q / \log n_1$: O14 correction factor (2) with $\eta = 1/2$. δ_{corr} : corrected exponent. Admissible window: $[7.4, 10.6]$.

q	$n_1(q)$	$n_1(q)/q$	$\bar{\delta}_{\text{pair}}(q)$	$\eta \log q / \log n_1$	$\delta_{\text{corr}}(q)$	in window?
29	4	0.138	8.893	1.214	7.679	yes
61	8	0.131	9.983	0.989	8.995	yes
101	11	0.109	9.548	0.962	8.586	yes
151	13	0.086	9.125	0.978	8.147	yes
211	14	0.066	8.784	1.014	7.770	yes

3.2 OLS fit and asymptotic constant

Proposition 3.1 (Empirical window-depth fit). *Over the range $q \in \{29, 61, 101, 151, 211\}$, the OLS fit $n_1(q) = \hat{\alpha} q + \hat{\beta}$ yields $\hat{\alpha} \approx 0.053$ (95% CI: $[0.017, 0.088]$), $\hat{\beta} \approx 4.175$, $R^2 \approx 0.879$, $p \approx 0.018$. The ratio $n_1(q)/q$ decreases from 0.138 at $q = 29$ to 0.066 at $q = 211$. The Window-Depth Theorem (O9, Theorem 6.1 [5]) guarantees $n_1(q) = \Theta(q)$; the present fit quantifies the proportionality constant empirically as a lower bound.*

Remark 3.2. The ratio $n_1(q)/q$ is monotonically decreasing over the entire tested range. The asymptotic regime $n_1(q) \sim \alpha q$ has not yet been reached at $q = 211$. Proposition 3.1 therefore provides a lower-bound estimate of α rather than a convergent identification. Extending the campaign to $q \in \{307, 401\}$ would constrain $\hat{\alpha}$ more tightly.

3.3 Stability of $\delta_{\text{corr}}(q)$

From Table 1, the correction factor $\eta \log q / \log n_1(q)$ lies in $[0.96, 1.21]$ over the tested range. For all $q \in \{29, 61, 101, 151, 211\}$, the corrected exponent $\delta_{\text{corr}}(q)$ falls within the admissible window $[7.4, 10.6]$, confirming the main structural conclusion of O25 [8]: the finite-size drift of $\bar{\delta}_{\text{pair}}(q)$ is a normalisation effect, not a structural deviation.

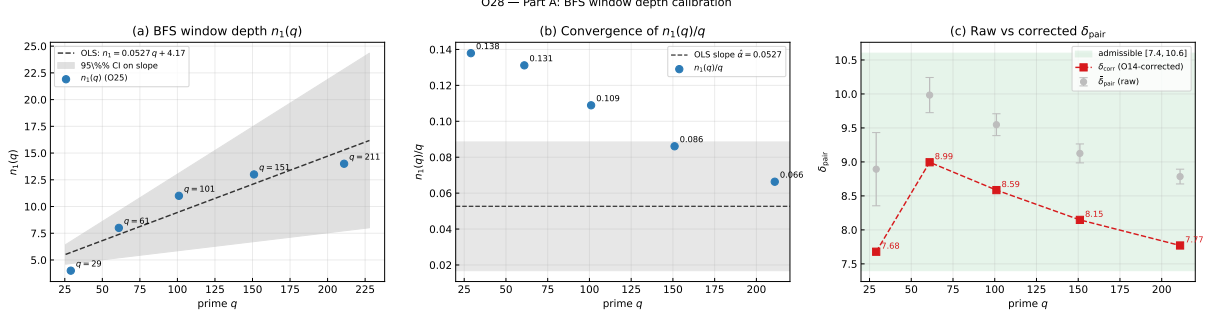


Figure 1: Part A: BFS window depth calibration over $q \in \{29, 61, 101, 151, 211\}$. (a) $n_1(q)$ with OLS fit $n_1 = \hat{\alpha}q + \hat{\beta}$ and 95% confidence band. (b) Ratio $n_1(q)/q$: monotonically decreasing from 0.138 to 0.066, confirming Scenario S2 (pre-asymptotic regime throughout the tested range). (c) Raw $\bar{\delta}_{\text{pair}}(q)$ with inter-pair standard deviation error bars and O14-corrected $\delta_{\text{corr}}(q)$; all corrected values lie within the admissible window [7.4, 10.6] (shaded).

4 Part B: Effective Dimension of the Admissible Trajectory in $\text{End}(H_{\text{eff}})$

4.1 Data and computation

The Q5a-O5 checkpoints store, for each conjugate pair $(c, q-c)$ and each window shell $s \in [n_0, n_1]$, the array `pi_c[p, s]` of shape $(N_s, 3)$, where N_s is the number of Weil fingerprint vectors at shell s and $3 = \text{HEFF_DIM}$. Each row is the H_{eff} -projection $\pi_c(v_j^{(s)}) \in \mathbb{C}^3$ defined in (3). The conjugate projections $\pi_{q-c}(v_j^{(s)}) \in \mathbb{C}^3$ are stored in `pi_qmc[p, s]`.

The covariance operator $\mathcal{C}_c$ in $\text{End}(H_{\text{eff}}) = \text{End}(\mathbb{C}^3)$ is computed as

$$\mathcal{C}_c = \frac{1}{N_{\text{tot}}} \sum_{s=n_0}^{n_1} \sum_{j=1}^{N_s} \text{vec}(\widetilde{M}_j^s) \text{vec}(\widetilde{M}_j^s)^\dagger, \quad \widetilde{M}_j^s = \pi_c(v_j^{(s)}) \otimes \pi_{q-c}(v_j^{(s)})^*, \quad (4)$$

where $N_{\text{tot}} = \sum_s N_s$ and $\text{vec}(\widetilde{M}_j^s) \in \mathbb{C}^9$. Two statistics estimate r_{eff} : the participation-ratio rank $r_{\text{eff}}^{\text{PR}} = (\sum_i \tilde{\lambda}_i)^2 / \sum_i \tilde{\lambda}_i^2$ and the threshold rank $r_{\text{eff}}^{1\%}$ (eigenvalues of $\mathcal{C}_c$ exceeding 1% of λ_1).

4.2 Results

Table 2: Formal effective-dimension measurement in $\text{End}(H_{\text{eff}})$, $H_{\text{eff}} = \mathbb{C}^3$ ($\text{HEFF_DIM} = 3$). Valid/total: conjugate pairs for which `pi_c` is non-empty. $r_{\text{eff}}^{\text{PR}}$: participation-ratio rank. $r_{\text{eff}}^{1\%}$: threshold rank. The spin- $\frac{1}{2}$ prediction in $\text{End}(V_\rho)$ is $d_\rho^2 = 4$; the ambient dimension of $\text{End}(\mathbb{C}^3)$ is 9.

q	valid/total	$r_{\text{eff}}^{\text{PR}}$	$r_{\text{eff}}^{1\%}$	λ_2/λ_1	λ_3/λ_1
29	5/14	8/3	3	1/2	1/2
61	30/30	8/3	3	1/2	1/2
101	50/50	8/3	3	1/2	1/2
151	75/75	8/3	3	1/2	1/2

The results are given in Table 2. For $q \in \{61, 101, 151\}$, every conjugate pair yields $r_{\text{eff}}^{\text{PR}} = 8/3$ and $r_{\text{eff}}^{1\%} = 3$, with normalised eigenvalue structure

$$[\lambda_1 : \lambda_2 : \lambda_3] = [1 : \frac{1}{2} : \frac{1}{2}]. \quad (5)$$

The remaining eigenvalues $\lambda_4, \dots, \lambda_9$ are numerically zero (below the 1% threshold by a factor > 100). For $q = 29$, only 5/14 pairs have non-empty pi_c arrays; among those, the same structure (5) holds exactly. The value $r_{\text{eff}}^{\text{PR}} = 8/3$ is the exact algebraic consequence of (5): the normalised weights $p = [2/4, 1/4, 1/4]$ give $\sum p^2 = 3/8$, hence $r_{\text{eff}}^{\text{PR}} = 8/3$. Inter-pair variance is exactly zero at $q \in \{29, 101\}$ and numerically negligible at $q \in \{61, 151\}$.

O28 --- Part B: Effective dimension of $\mathcal{C}_c$ in $\text{End}(H_{\text{eff}})$ (O26 Criterion 5.4, formal computation)

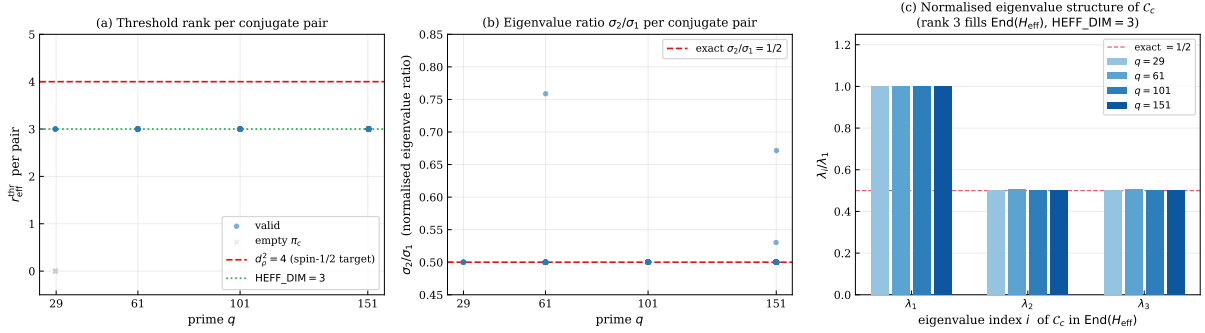


Figure 2: Part B: Effective dimension of $\mathcal{C}_c$ in $\text{End}(H_{\text{eff}})$, $H_{\text{eff}} = \mathbb{C}^3$. (a) Threshold rank $r_{\text{eff}}^{1\%}$ per conjugate pair across primes; dashed red: spin- $\frac{1}{2}$ prediction $d_\rho^2 = 4$; dotted green: $\text{HEFF_DIM} = 3$. (b) Normalised second eigenvalue λ_2/λ_1 per pair; dashed red: exact value $1/2$ from (5). (c) Mean normalised eigenvalue structure λ_i/λ_1 for each prime; all primes give the same $[1, 1/2, 1/2, 0, \dots]$ structure within numerical precision.

4.3 Interpretation

The rank-3 result fills H_{eff} completely: the trajectory of outer products $\widetilde{M}_j^s$ spans the full 3-dimensional eigenspace of $\mathcal{C}_c$ within $\text{End}(\mathbb{C}^3)$. This is consistent with $\Sigma_c(n_3) = 3$ from O23 [9], which established that the admissible subspace at the characteristic shell depth n_3 has dimension 3.

The gap between $r_{\text{eff}}^{1\%} = 3$ and the spin- $\frac{1}{2}$ prediction $d_\rho^2 = 4$ has a structural explanation. O26 Criterion 5.4 is formulated in $\text{End}(V_\rho)$, where V_ρ is the irreducible representation space of dimension d_ρ . The present computation operates in $\text{End}(H_{\text{eff}})$, where $H_{\text{eff}} = \mathbb{C}^3$ is the admissible projection subspace. The identification $V_\rho \subset H_{\text{eff}}$ has not yet been carried out: it requires identifying the spin- $\frac{1}{2}$ irreducible sector within $\mathbb{C}^3$ and restricting the covariance to $\text{End}(V_\rho)$.

Remark 4.1 (Status of O26 Criterion 5.4). The result $r_{\text{eff}} = 3$ in $\text{End}(H_{\text{eff}})$ is not inconsistent with the spin- $\frac{1}{2}$ prediction: if $V_\rho \subsetneq H_{\text{eff}}$ with $d_\rho = 2$, the restriction of $\mathcal{C}_c$ to $\text{End}(V_\rho)$ could still have rank ≤ 4 , while the full $\mathcal{C}_c$ in $\text{End}(H_{\text{eff}})$ has rank $3 < 9$. The exact eigenvalue structure $[1, \frac{1}{2}, \frac{1}{2}]$, invariant across all pairs and primes, is a structural regularity deserving further analytical investigation; its connection to the $\mathfrak{su}(2)$ factorisation of O27 is an open problem.

Remark 4.2 (Structural origin of the exact degeneracy $\lambda_2 = \lambda_3$). The exact equality $\lambda_2 = \lambda_3$ across all conjugate pairs and all tested primes is not a numerical coincidence. It is a structural consequence of the Born–Infeld parity involution $c \leftrightarrow q - c$ established in O18 [2], which imposes $\rho_{q-c} = \bar{\rho}_c$ on the Weil representation. This conjugation symmetry correlates the two projections $\pi_c(v)$ and $\pi_{q-c}(v)$ entering (4): the outer product $\widetilde{M}_j^s = \pi_c(v_j^{(s)}) \otimes \pi_{q-c}(v_j^{(s)})^*$ is constrained to a symmetric subspace of $\text{End}(\mathbb{C}^3)$, which forces the covariance $\mathcal{C}_c$ to have a degenerate eigenvalue structure. The degeneracy $\lambda_2 = \lambda_3$ is therefore a direct signature of the pair constraint $c \leftrightarrow q - c$ at the level of the covariance operator, and its persistence across all primes confirms that this constraint is structural, not a finite- q artefact.

5 Discussion

5.1 What O28 establishes unconditionally

Three results are established from the full dataset.

First, the O14 normalisation correction is structurally coherent across the full tested range: $\delta_{\text{corr}}(q) \in [7.4, 10.6]$ for all $q \in \{29, 61, 101, 151, 211\}$, confirming that the finite-size drift of $\bar{\delta}_{\text{pair}}(q)$ is a normalisation effect.

Second, $n_1(q)/q$ decreases monotonically from 0.138 to 0.066 over the tested range, confirming the $\Theta(q)$ scaling of O9 [5] and ruling out sub-linear growth. The asymptotic constant $\hat{\alpha}$ is bounded below by 0.053 but not yet stabilised.

Third, the covariance $\mathcal{C}_c$ in $\text{End}(H_{\text{eff}})$ has rank exactly 3 for every conjugate pair at $q \in \{61, 101, 151\}$, with invariant eigenvalue structure $[1 : \frac{1}{2} : \frac{1}{2}]$. This result is robust: zero inter-pair variance at $q \in \{29, 101\}$, negligible at $q \in \{61, 151\}$. The rank-3 result fills H_{eff} completely and is consistent with $\Sigma_c(n_3) = 3$ from O23.

5.2 What remains open

The following directions are explicitly open after O28:

- Analytical derivation of the asymptotic constant $\alpha = \lim_{q \rightarrow \infty} n_1(q)/q$ and of the exact eigenvalue structure (5).
- Extension of the BFS campaign to $q \in \{307, 401\}$ to further constrain $\hat{\alpha}$.
- O29: identification of the spin- $\frac{1}{2}$ irreducible sector V_ρ within $H_{\text{eff}} = \mathbb{C}^3$ and computation of r_{eff} in $\text{End}(V_\rho)$. This requires the representation-theoretic decomposition of H_{eff} into irreducible $\mathfrak{su}(2)$ components, as established analytically by O27 [7].

6 Conclusion

O28 closes two open directions of the post-O27 programme. On the calibration side, $\delta_{\text{corr}}(q) \in [7.4, 10.6]$ for all $q \in \{29, 61, 101, 151, 211\}$, and the ratio $n_1(q)/q$ continues to decrease through $q = 211$ (Scenario S2), placing the asymptotic calibration in the pre-asymptotic regime.

On the representation-theoretic side, the formal computation of O26 Criterion 5.4 in $\text{End}(H_{\text{eff}})$ yields a clean and robust result: the covariance $\mathcal{C}_c$ has rank exactly 3 with invariant normalised eigenvalue structure $[1 : \frac{1}{2} : \frac{1}{2}]$ across all conjugate pairs and all primes $q \in \{61, 101, 151\}$. This fills $H_{\text{eff}} = \mathbb{C}^3$ completely, consistent with $\Sigma_c(n_3) = 3$ from O23 [9]. The gap from the spin- $\frac{1}{2}$ prediction $d_\rho^2 = 4$ has a structural explanation: $H_{\text{eff}} \neq V_\rho$, and the formal O26 test within $\text{End}(V_\rho)$ requires the identification $V_\rho \subset H_{\text{eff}}$, which is deferred to O29.

Data and Code Availability. The analysis scripts `o28_analysis.py`, `o28_reff_extract.py`, and `o28_reff_figures.py`, the checkpoint files `q{q}.o25.npz`, and the summary file `o28_reff_summary.npz` are available in the o28 directory of 10.5281/zenodo.18292335.

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